



Six Years of Shiny in Research - Forging Innovative Methodologies

Owen Horton¹, Zoe Doherty²

Introduction

- Shiny simplifies the use of complex methodologies for researchers of different specialties, and its free and simple implementation has sparked popularity among collaborators in a variety of fields.
- However, Shiny users without a solid background in data analytics may cause irreproducibility in peer-reviewed literature, and Shiny app developers without a foundational understanding of data security may not understand web internet protocols of transferring data securely
- So to what extent has Shiny provided knowledge during its six years of existence? An algorithmic analysis on Shiny-related academic literature and a discussion of underlying limitations aim to answer that question for the everyday Shiny user

Algorithms/Methods for Literature Search

A literature search was conducted using Web Scholar, Scopus, Google Scholar, and University of Adelaide (UofA), with primary keywords “Shiny Web Application” or “Shiny Web App”. The purpose was to isolate a subset of academic literature capable of offering a substantive and holistic reflection of Shiny’s role in research. Pulled from at least one of the four search tools, each of ~127,000 papers from 2012-2018 were given a relevance score from the Google Scholar algorithm, which depends on a record’s degree of keyword match, academic significance, type of search, and publication date.

The top 600 highest-scoring papers were selected for thorough reading, as results following them in relevance often only cited its use as a tool. Among these selected papers, 445 original applications were introduced, 373 unique R packages were utilized, and 229 unique peer-reviewed journals were represented.

The papers with high relevance score generally discussed Shiny specifically and featured “Shiny” in the title or the subject fields, increasing their relevance score. Earlier papers were more likely to discuss Shiny, with newer papers more likely to mention Shiny in the text only.

Technical Aspects

Shiny’s technical aspects harbor unique benefits and limitations:

- Hosting:** Shiny is free for 5 applications and 25 active hours, but grows exponentially expensive otherwise. For those with an in-depth knowledge of internet security, it is more economical to host the application independently (Core Team, 2012).
- Security:** In web security, a developer cannot build defenses for attack vectors they are unfamiliar with. It is generally wise to leave the specifics of data security to experts in the field (Charpentier, 2013).
- Architectural Structure:** Shiny actively encourages the use of single-file applications but this structure falls apart as an application grows and increases in complexity (Chang et al., 2018).
- Data Size:** Shiny has no inbuilt data streaming functionality, forcing the application to require pre-partitioned uploads (R Core Team, 2017).

Literature Analysis

- Results spanned **229 unique peer-reviewed journals** across diverse disciplines with Bioinformatics publishing the largest share.
- Areas lower in data analysis expertise (e.g. Biological Sciences, Education) showed notably **high Shiny usage**
- Shiny is primarily used as a **visualization and dissemination tool**, with the most substantive discussions appearing in earlier papers.

Four Major themes emerged (Kasprzak et al., 2020):

- Complexity barrier:** Shiny lowers the barrier for non-programmers to use advanced methods
- Cross-collaboration:** Apps bridge skill gaps between field researchers, theorists, and practitioners
- Flexibility:** Shiny links specialized data gathering tools to existing software, avoiding costly manual data transfers.
- Reproducibility:** Shiny-wrapped code aids transparent, reproducible research
- Education:** Shiny has proven effective as a teaching tool for statistical concepts

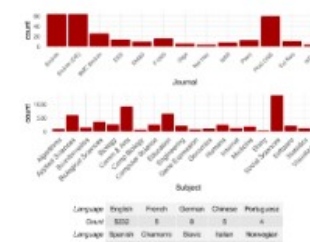


Figure 1: (Left) word counts among Shiny publications by journal, subject, and language (Kasprzak et al., 2020)

- Figure 1 shows Shiny’s wide disciplinary reach, spanning journals from Bioinformatics to Social Science (Kasprzak et al., 2020)

Conclusion

- Shiny is not a silver bullet, but it enables **specialized applications** created cheaply and easily for any level of end user (Kasprzak et al., 2020)
- Its greatest contribution is enabling **cross-disciplinary collaboration**, allowing field researchers, clinicians, and educators to engage with cutting-edge methods without programming expertise.
- Data security** is a major concern - web protocols are a non-trivial skill set for many R programmers.
- Architectural limitations such as lack of debugging tools and single-file encouragement will **limit complexity** of future applications.
- Shiny promotes **open and reproducible research** and offers a real pathway to making complex methodologies accessible outside of research.

References

Kasprzak, et al., “Six Years of Shiny in Research - Collaborative Development of Web Tools in R”, The R Journal, 2021